2025_Hexaware_Actual_Test_6

Test Summary

- No. of Sections: 2
- No. of Questions: 90
- Total Duration: 90 min

Section 1 - Aptitude

Section Summary

- No. of Questions: 60
- Duration: 60 min

Additional Instructions:

None

Q1. If 3664XX is divisible by 36. Find the value of X^2?

- 64
- 16
- 4
- 36

Q2. Find out the H.C.F of 3^5 , 3^7 and 3^{11}

- 9
- 243
- 81
- None of the mentioned options

Q3. A boy bought a camel and carriage for Rs. 5000. He sold the camel at a gain of 20% and the carriage at a loss of 10%. If he gains 3% on the whole, then find the cost of the camel.

- 2166.67
- 2178.37
- 2138.68
- 2159.94

Q4. A shopkeeper bought an item at Rs. 3600 and while selling it offered a 40% discount on the marked price and earned a profit of Rs.1200. If a discount is reduced by 10%, then what is the percentage profit on the cost price?

- 34.44%
- 42.22%
- 48.88%
- 38.88%

Q5. If 10% of Guru's income = 15% of Raj's income = 20% of Kamal's income. If the sum of their income is 3900, find the income of Kamal.

- 1200
- 900
- 1800
- 600

Q6. At an election between two candidates, the candidate who got 66 % of the votes casted won by 3072 votes. Find the total number of the voters on the voting list if 80% people casted their vote and there were no invalid votes.

- 15000
- 12000
- 14000
- 18000

Q7. A sum of money was invested at SI at a certain rate for 3 years. Had it been invested at a 4% higher rate, it would have fetched Rs. 480 more. Find the principal.

- Rs. 4000
- Rs. 4400
- Rs. 5000
- Rs. 3500

Q8.

If the simple interest on a certain sum of money is 4/25 of the sum and the rate per cent equals the number years, then the rate of interest per annum is:

- 4%
- 5%
- 8%
- 10%

Q9. 30 women completes a piece of work in 90 days working 8 hours/day. Find the number of working hours taken by 24 men works 100 days to complete the same work. (Efficiency of man and woman is equal)

- 9 hours
- 8 hours
- 10 hours
- 12 hours

Q10. If 3 men with 4 boys can earn 2100 in 7 days and 11 men with 13 boys can earn Rs. 8300 in 8 days, In how days will 7 men with 9 boys earn Rs. 11000?

- 18 days
- 14 days
- 16 days
- 20 days

Q11. A train covered first 120 km at a speed of 20 km an hour and then covered the remaining 180 km at a speed of 45 km an hour. Find its average speed.

- 20km/hr
- 30km/hr
- 10km/hr
- 25km/hr

Q12. Vinu takes as much time in running 30 meters as a car takes in covering 48 meters. What will be the distance covered by vinu during the time the car covers 2 km?

1.5 km

1.25 km

1.8 km

None of the above

Q13. The average age of workers in field and in documentation in a factory was 45 year. The average age of all the 16 documentation workers was 38 years and average age of field workers was 52 years. If 7 field workers were married then the number of unmarried field workers was:

6

7

8

9

Q14. The average of six numbers is 78. If the average of the first five numbers is 56, what is the value of the sixth number?

120

188

158

175

Q15. Abi and Kiran invested in a business. They earned some profit which they divided in the ratio of 3:2. If Kiran invested Rs. 20000. Find the amount invested by Abi.

30000

31500

14000

28000

Q16. Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?

16

18

10

12

Q17. In a city, there are few engineering, MBA, and CA candidates. The sum of four times the engineering, three times the MBA, and 5 times CA candidates is 3650. Also, three times CA is equal to two times MBA, and three times engineering is equal to two times CA. In total how many MBA candidates are there in the city?

200

300

450

400

Q18. A class is composed of two brothers and six other boys. In how many ways can all the boys be seated at a round table so that the two brothers are not seated beside each other?

720

1440

3600

4320

Q19. From 10,000 lottery tickets numbered from 1 to 10,000, one ticket is drawn at random. What is the probability that the number marked on the drawn ticked is divisible by 20?

1/10

1/20

3/20

1/40

Q20. A die is thrown twice. The probability that at least one of the two throws comes up with the number 3 is:

5/12

11/36

1/12

7/24

Q21. **Directions:**
What will replace the question mark (?) in the following series?

1, 3, 9, 39, 249, 2559, ?.

Q22. **Directions:**
What will replace the question mark (?) in the following series?

2, 10, 30, 68, 130, ?.

Q23. **Directions:**
What will come in place of question mark (?) in the following series?

54, 225, 576, 729, ?, 9.

Q24. Pointing to a man in the photograph, a woman said, “His brother’s father is the only son of my grandfather.” How is the woman related to the man in the photograph?

Mother

Aunt

Sister

Daughter

Q25. When Martin saw Hannah, he recalled, “She is the daughter of the father of my daughter’s mother.” Who is Hannah to Martin?

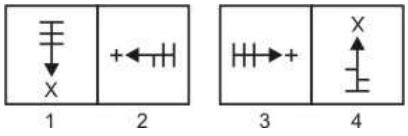
Sister-in-law

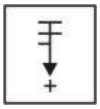
Sister

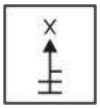
Cousin

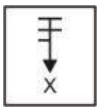
Aunt

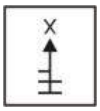
Q26. **In the question, two pairs of problem figures marked 1, 2, 3, and 4 and four answer figures are given. Select the best answer figure to substitute element 4 in the problem figures so that element 3 is related to element 4 in the same way as element 1 is related to element 2. To establish the relationship, consider maximum aspects of the figures (i. e., size, shape, movement, etc.). The number of that answer figure which can be substituted for problem figure 4 is your answer.**



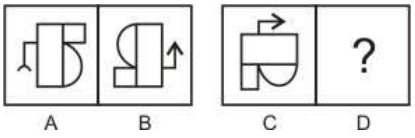


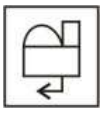






Q27. The second figure in the first part of the problem figures bears a certain relationship to the first figure. Similarly, one of the figures in the answer figures bears the same relationship as the first figure in the second part. You have to select the figure from the set of answer figures that would replace the sign of the question mark (?).











Q28. **In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.**

Statement: X≤Y<W =Z ≤U<S;S>T ≥V

Conclusions:

I. Z≥T
II. Z > X

- if only conclusion 1 is true
- if only conclusion 2 is true
- if either conclusion 1 or conclusion 2 is true
- if neither conclusion 1 nor conclusion 2 is true
- if both conclusion 1 and conclusion 2 are true

Q29. In the question below relationship between different elements is shown in the statements. The statements are followed by two conclusions.

Statement: L ≥ M ≤ R = S; M > N ≥ P

Conclusions:

I. P ≤ M

II. L > N

- if only conclusion I is true
- if only conclusion II is true
- if either conclusion I or conclusion II is true
- if neither conclusion I nor conclusion II is true
- if both conclusion I and conclusion II are true

Q30. The question below consists of a question and two statements numbered A and B given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both the statements and give an answer.

In how many days will the two machines A and B produce 500 widgets if they produce them in equal numbers?
(A) Machine A alone can produce 500 widgets in 10 days.
(B) Machine B is 30% faster.

- if the question can be answered by one of the statements alone, but cannot be answered by using the other statement alone
- if the question can be answered by using either statement alone.
- if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.
- if the question cannot be answered even by using both statements together.

Q31. The question below consists of a question and two statements numbered I and II given below it. You have to decide whether the data provided in the statements are sufficient to answer the questions. Read both the statements and give answer:

In a certain code, nop all ed' means 'They like flowers'. Which code word means 'flowers'?
I. 'id nim nop' means 'They are innocent'.
II. 'gob ots all' means 'We like roses'.

- If the data in statement I alone are sufficient to answer the question, while the data in statement II alone are not sufficient to answer the question
- If the data in statement II alone are sufficient to answer the question, while the data in statement I alone are not sufficient to answer the question
- If the data given in both statements I and II together are not sufficient to answer the question
- If the data in both statements I and II together are necessary to answer the question

Q32. **In the question given below is given a statement followed by two conclusions numbered I and II. You have to assume everything in the statement to be true, then consider the two conclusions together and decide which of them logically follows beyond a reasonable doubt from the information given in the statement**

Statements: Some doctors are fools. Some fools are rich.

Conclusions:
I. Some doctors are rich.
II. Some rich are doctors

- If only conclusion I follows.
- If only conclusion II follows
- If either I or II follows.
- If neither I nor II follows.

Q33. **Given two statements, verify the conclusions and mark the answer**

Statements:
All Grapes are Bananas.
All Bananas are Potatoes.

Conclusions:
I. Some Potatoes are Bananas.
II. Some Grapes are Potatoes

- if only conclusion I follows.
- if only conclusion II follows.
- if both conclusions I & II follow.

if no conclusion follows.

Q34. **Fill in the blank with a suitable article from the options given.**

We had _____ really good time in Tirupathi.

a

an

the

no article required

Q35. **Fill in the blanks by choosing the right preposition:**

My sister is very angry _____ me for breaking her toy.

on

upon

at

in

Q36. **Complete the following sentence by choosing the right option.**

_____ this kid, I found a great thirst _____ learning.

Through, along with

About, for

In , in

In , for

Q37. **Identify the part which has error in it:**

The creature on Mars, if any, (A) / are bound to be very different from us (B) / not only in shape but also in size (C) / because of different gravitational conditions (D).

A

B

C

D

Q38. **Read each sentence to find out whether there is any error in it. The error, if any, will be in one part of the sentence. The number of that part is the answer.**

Anybody could (1) meet the president (2) if they (3) wished to (4)

1

2

3

4

Q39. **Choose the word that best fills the blank preserving the meaning of the sentence as a whole.**

Ironically, Nirmal Verma's precision in sketching lives that are on the edge of despair ensures that his stories will sometimes be read too narrowly, much as Premchand's social-reformer role once caused his broader concerns to be _____

ignored

reinforced

contradicted

diminished

Q40. **In the question given below, a sentence is given with two blanks, followed by five options, each having two words that may or may not fit in the blanks. Choose the one that gives the correct words that fit the blank.**

A _____ of China's human rights record and its challenge to American geopolitical leadership may _____ viewer enthusiasm in the crucial American market.

Faction, reduce

Historical, diminish

Accord, abate

Combination, dent

Q41. **The sentences given in each question, when arranged in the proper sequence, form a coherent paragraph. Each sentence is labeled with a letter. Choose the most logical order of sentences from among the given choices to construct a meaningful paragraph.**

(A) Yet few of the 500000 residents of the former Spanish colony are feeling flush with petrodollars; most have no plumbing or electricity.
(B) Although the coup failed, much remains at stake in the shaky oil state, where offshore drilling began in the early 1990s.
(C) "None of the oil revenues have gone into developing the country", says Sarah Wykes, a researcher for the London corporate watchdog group Global Witness.
(D) Production of about 250000 barrels per day is projected to nearly double in the next few years.

- CADB
- DBAC
- BDAC
- BDCA

Q42. **Directions :**
In the following question, a sentence is given with certain words marked as (A), (B), (C), (D), and (E). Choose the option with the sequence of sentences that makes the paragraph grammatically and contextually correct.

(A) There are many tests to screen and diagnose cancer but the most common one used is by examination of a biopsy sample of the suspected cancer tissue.
(B) Cancer is a disease that is caused when cells develop abnormally anywhere in the body these growths can be
(C) The cancer spread is generally given a number from 0 to 4 the last being the most aggressive.
(D) fatal, but some of them are harmless, there are over Two hundred types of cancer that have prevailed over the years.
(E) Determining the stage of cancer also helps in determining treatment protocols.

- (a) DBCEA
- (b) BAEDC
- (c) BDACE
- (d) CADEB

Q43. **Fill in the blank with the right verb form:**

Which books _____to college yesterday?

- did you take
- do you take
- have you taken
- took you

Q44. **Fill in the blank with the right option:**

You better not _____to keep the keys in the right place.

- forgetting
- are forgetting
- forgotten
- forget

Q45. **From the given options choose the one that best expresses the given sentence in active/passive voice:**

I will never play this game

- This game is not played by me.
- This game would never be played by me.
- This game will never be played by me.
- None of the mentioned options

Q46. **Change from direct speech to indirect speech**

"I've been working for the same company since 1960," he said to me.

- He told me that he is working for the same company since 1960.
- He told her that he had been worked for the same company since 1960.
- He told me that he was working for the same company since 1960.
- He told me that he had been working for the same company since 1960.

Q47. **Select the most suitable synonym**

yokel

- rustic
- ignoramus
- dandy
- recluse

Q48. Select the most suitable synonym.

Judicious

fair

honest

prudent

loyal

Common Content:

Kindly study the following information carefully and answer the question that follows:

In a family of three generations, there are eight members - A, B, C, D, E, F, G and H. C is G's daughter-in-law. E is B's only grandchild. D is one of E's two maternal uncles. G and D, one of whom is male, are the children of B. F is the father of D. H is G's husband.

Q49. How is E related to F?

Granddaughter

Son

Grandson

Daughter

Cannot be determined

Q50. Which of the following statements is true?

There are two males in the second generation.

A is the sister-in-law of H.

There are equal number of males and females in the family.

C is the granddaughter in law of F.

Common Content:

B, D, H, J, P, S and T are seven students studying in seven different schools i.e. St.Anne's, St.Joseph, Sri Amrita, Nava Bharath, KendriyaVidyalaya, Cauvery Global and BalaVidyaMandir, but not necessarily in the same order.
They have a special class on one day from Monday to Sunday. No two students have classes on the same day.

1. S is studying in Cauvery Global and he does not have special class on Friday or Wednesday.
2. The one who study in Kendriya Vidyalaya has special class on Tuesday.
3. H is studying in BalaVidyaMandir and has a special class on Sunday.
4. D is studying neither in Nava Bharath nor in Sri Amrita.
5. T and P do not study in St.Anne's and do not have classes on Wednesday.
6. The one who study in St.Anne's is have a class immediately after P.
7. B is studying in St.Joseph and have a class on Monday.
8. There is only one class between B and One who study in Amrita.
9. T does not study in Nava Bharath.
10. D has a class on neither Thursday nor Friday.

Q51. Who among the following is studying Kendriya Vidyalaya School?

- J
- P
- T
- D

Q52. P is studying which school and he has a special class on which day?

- Nava Bharat – Friday
- Nava Bharat – Tuesday
- Sri Amrita – Tuesday
- Kendriya Vidyalaya – Friday

Q53. Which of the following combination is true?

- St.Anne's – H
- Bala Vidya Mandir – T
- St.Joseph's- B
- Nava Bharath –D

Q54. Student T belongs to

- Cauvery Global
- Nava Bharat
- Sri Amrita
- Kendriya Vidyalaya

Q55. Who among the following have a special class on Wednesday?

- P
- D
- J

cannot be determined

Common Content:

Read the passage and answer the questions that follow.

We should make a distinction between foundations and principles, but not assume that they are wholly separate. The principles of good governance are derived from the foundations of humans which, in order to be truly universal, must themselves be established in the sacredness of life, and not just in some intellectual notion of what human order is. Principles that are derived wholly from mental constructs are, like them, shifting and like them, as easily abandoned. When they are replaced by other principles, the latter are constructs no less, and the cycle is repeated. It is true that each began with a deep concern for human progress, but it is also true that the thing was not long before put into jeopardy. That is because, perhaps, the successive ideas of human progress were so fragmented and limited to one's wrongly understood self-interest that they never became truly universal. Whatever else the history of government based on the principles of utilitarianism, liberalism and socialism shows, it shows that most of all. For it brought up again and again and always with reference to the actual living conditions of the people, the question of whose progress?

Dharma demonstrates how human rights, even before they are a matter of principle, are a matter of foundation - of a sane society. It will not be circular reasoning at all if it is said that in actual fact, they are a matter of Dharma. The literal meaning of the Sanskrit word 'dharma' is the foundation. The foundations or Dharma, of all relationships, personal and social, lie in the recognition that life is sacred, that all its expressions are inter-related and bound in visible unity, and that no violence shall be done to a person by another except with the due process of law, which at all times must be free from anger and hatred, and that the political and legal governance of human relationships shall have as their chief aim the enhancement of the individual and the progress of the people. It is from these that the principles of law and government must be derived, and it is in them that they shall have their ultimate validation.

This is the Dharmic law. Its concern with the dignity and worth of the individual and with social progress was given institutional forms of jurisprudence. That was done in the Mahabharata and some centuries later in the Arthasastra of Kautilya. Both are of great relevance to our times. But here let us have a brief look at the latter of the two, assigned to 321 B.C. in the reign of Chandragupta Maurya, and the discovery, of which, in 1905 by R.Shamasastry, was an event of very great excitement. Ahimsa, which I prefer to translate as "non-violence," the foundation of social existence, leads to the first principle of the law of punishment or danda-niti. It is stated at the very beginning of the Arthasastra."Whoever imposes severe punishment becomes repulsive to the people, while he who awards mild punishment becomes contemptible. But whoever imposes punishment as being deserved becomes respectable. For punishment, when awarded with due consideration, makes the people devoted to dharma and to works productive of wealth and joy; while punishment, when ill-awarded under the influence of greed and anger owing to ignorance, excites fury even among the heretics and ascetics dwelling in forests, not to speak of householders."

This principle then became the basis of the law against illegal confinement of any person, man or woman, and of the law against the unauthorised criminal trial of anybody. While in the custody of the police or of the jail superintendent, no accused or convict shall be insulted and humiliated by offensive language, nor beaten or tortured. Moreover, the principle of "punishment as deserved" must at all times be coupled with the principle of mercy. In such circumstances, for example, a diseased person anyone suffering from hunger, thirst, or fatigue from journey, a villager from country parts, anyone that has suffered much from punishment, and a moneyless pauper shall be shown mercy."In brief, the Arthasastra principle is that even an accused and a convict is entitled to his or her dignity. Whoever official will violate these principles, shall himself be liable to punishment.

In the case of a woman, her ill-treatment or rape by an officer, in whose custody she was placed either in the lock-up or in the prison was regarded as a serious offence and the officer was punished, taking into account the relative degree of the offence, either by fine or even by death. The Arthasastra further prescribed that if a woman was pregnant, or had just delivered a child, the sentence on her was to be kept in suspension until she was in a fit condition to be subjected to it.

The evident truth that woman is the spring of life, and her personality was not to be violated if life were to be regarded as sacred, led to the principle - "No man shall have sexual-intercourse with any woman against her will." Kautilya repeats this three times in a single chapter, and it clearly applied to the man-woman relationship in marriage as well. Whatever might be the position of the sastras of the medieval times on the question of divorce, the Arthasastra is perfectly clear that both man and woman have equal status in that."A woman, hating her husband, cannot dissolve her marriage with him against his will. Nor can a man dissolve his marriage with his wife against her will. But from mutual enmity, divorce may be obtained."

But the greatest significance is the Arthasastra's admonition to the king, now the state - "In the happiness of his subject lies his happiness, in their welfare, his welfare, he shall not consider as good to what pleases himself, but to what please his subjects. It was to this supreme maxim that all his public acts were subjected. And because dharma as justice is the foundation of civil society, then it led to the principle that "the king shall with adequate punishments test first the conduct of government servants, and then only will examine the conduct of his people in towns and villages." For nothing is more destructive of social order than the evidence that the officers of the state are above the law.

For that reason, Kautilya places the conduct of judges under a discipline of the highest kind. To protect the people from unjust punishments, he prescribes the following. "When a judge threatens, browbeats, sends out or unjustly silences any one of the disputants in his court, if he does not ask what ought to be asked, or asks what ought not to be asked, leaves out what he himself has asked, makes unnecessary delay in, discharging his duties, compels the parties to leave the court by tiring them with delay, evades or causes to evade

statements that lead to the settlement of a case, he shall be punished with the highest amercement. If he repeats the offence, he shall both be punished with double the above fine-and dismissed."

Q56. In 'Arthasastra' greatest prominence is given to

- the maxim that his children's welfare is the king's welfare.
- dharma as the foundation of social organization.
- the principle that people's happiness is the Kings's happiness.
- the waning or reproof by the king.

Q57. No violence should be perpetrated upon one person by another except with a proper system of law which

- should aim at heightening of the individual.
- must be free from anger and hatred.
- should look towards progress of the people.
- should have final confirmation.

Q58. The principles of good administration originate from the foundations of human order but they also should evolve from

- some empirical idea.
- the sanctity of life.
- some intellectual idea.
- the mental constructs.

Q59. 'Dharma' shows that human rights, before becoming principles, make the foundation of a sensible society. Following is/are the intricacies of this phenomenon.

- the meaning of Dharma is foundation.
- human rights are an issue of Dharma.
- the foundations of all relationships are in knowing the sanctity of life.
- all its relationships are interrelated.

Q60. Punishment awarded with one due care earns people's loyality to 'Dharma' and to creativeworks, when ill-awarded, earns

contempt

respect

wrath

criticsm

Section 2 - Pseudocode and Computer Fundamentals

Section Summary

- No. of Questions: 30
- Duration: 30 min

Additional Instructions:

None

Q1. Which of the architecture is power efficient?

RISC

ISA

IANA

CISC

Q2. What does CSA stand for?

Computer Service Architecture

Computer Speed Addition

Carry Save Addition

None of the mentioned options

Q3. The register which contains the data to be written into or read out of the addressed location is known as

Index register

Memory address register

Memory data register

Program counter

Q4. A compiler program written in a high-level language is called _____

Source Program

Object Program

Machine Language Program

None of the mentioned options

Q5. A collection of 8 bits is called _____.

byte

record

word

nibble

Q6. Communication circuits that transmit data in both directions but not at the same time are operating in

Simplex mode

Half-duplex mode

Full-duplex mode

Asynchronous mode

Q7. Firmware means

Software

Hardware

Software available on hardware

None of the mentioned options

Q8. Non-volatility is an important advantage of

- CCDs
- Magnetic tapes and disks
- Magnetic bubbles
- Both Magnetic tapes, disks and Magnetic bubbles

Q9. Updating a database means

- Revising a file structure
- Reorganizing the database
- Modifying or adding records
- All of the mentioned options

Q10. EEPROM is

- Easily erasable Programmable Read-Only Memory
- Non-erasable Programmable Read-Only Memory
- Effectively Erasable Programmable Read-Only Memory
- Electrically Erasable Programmable Read-Only Memory

Q11. Accumulator is a

- Hardwired unit
- Sequential circuit
- Finite state machine
- Register

Q12. Bar codes stores information using

- punched holes
- dots

- thick and thin lines
- all of the mentioned options

Q13. Which of the following programs enables you to calculate numbers related to rows and columns?

- Window program
- Spreadsheet program
- Word program
- None of the mentioned options

Q14. A computer is accurate, but if the result of a computation is false, what is the main reason for it?

- Power failure.
- Distraction.
- Incorrect data entry.
- All of the mentioned options.

Q15. Which of the following is the first neural network computer?

- AN
- AM
- SNARC
- RFD

Q16. What will be the output of the following pseudocode?

```
1 Integer a,b,c
2 Set a=7,b=3,c=10
3 a=(c+a)+b
4 if (c&b&a)<(b&a&c)
5    a=(1+2)+c
6 End if
7 Print a+b+c
```

- 44

28

37

33

Q17. What will be the output of the following pseudocode?

```
1  Integer a,b,c,d,e
2  Set a=4,b=12,c=41,d=0
3  while (a+b<c)
4     d=d+a+b+c
5     e=d/3
6     a=a+5
7     b=b+5
8  end while
9  Print e
```

3

67

111

None of the mentioned options

Q18. What will be the output of the following pseudocode?

[Note: MOD finds the remainder after the division of one number by another. For example, the expression "5 MOD" would evaluate to 1 because 2 leaves a quotient of 2 and a remainder of 1]

```
1  integer array1[6],p,j,q
2  set p=3
3  set array1[6]={3, 6, 10, 12, 23, 33}
4  for (each j from 0 to 5)
5  if ((array1 [j] MOD p) EQUALS 0)
6  p = array1[j] - (p * 3)
7  end if
8  q= p+ array1[j] - 3
9  end for
10 print q
```

64

44

34

54

Q19. What will be the output of the following pseudocode for c=1?

```
1  Integer fun(Integer c)
2     Print c
3     if(c<3)
4        c=c+2
5        fun(fun(c)
6     end if
7  return c
```

```
8  end function fun
```

1
3

1
3
3

3

None of the mentioned options

Q20. What will be the output of the following pseudocode?

```
1  Integer j, m
2  set m= 1
3  integer a[4] = { 1, 1, 1, 2}
4  a[2]= a[1] - a[2] + a[3]
5  a[3] = a[2]
6  m= a[3]
7  print m
```

-6

16

2

3

Q21. What will be the output of the following pseudocode for a= 22?

```
1  integer compact(integer a)
2  if( a <= 45)
3  compact( compact (compact (a * 5 + 1)))
4  end if
5  return a*2
6  end function compact ()
```

22

45

56

44

```
1  Integer a, b, c
2  Set a=5, b=7, c=7
```

```
3  for(each c from 3 to 5)
4    if((a+b+c)>(c-a))
5     b=(a&c)+c
6     a=c+c
7   End if
8   a=(b^c)+c
9  End for
10 Print a+b
```

- 24
- 26
- 28
- 30

Q23. What will be the output of the following pseudocode?

```
1  Integer p, q, r
2  Set p=7, q=8, r=7
3  r=(1^4) +r
4  p=(q+12)+r
5  q=(q+2)&p
6  if (((r&q)<q || 5>q)
7    p = r+p
8    p= (r+r)+p
9  End if
10 p=(q&9)+p
11 p=(r+r)+p
12 Print p+q+r
```

- 161
- 107
- 104
- 106

Q24. What will be the output of the following pseudocode for a=4 and b=2?

```
1  Integer funn(Integer a, Integer b)
2    if (a>b || b<a)
3       return 1+funn(a+a-b+b+b+a, a+a-b+b+b+a)
4    End if
5    return a
```

- 10
- 19

27

15

Q25. What will be the output of the following pseudocode?

```
Integer a=33,j=9;
for (each i from 1 to 3)
   b=j/i
   a=a+b
   if (a mod 2 EQUALs 0)
      Print a
   Else
      Print a-1
End for
```

34
36
38

42
46
48

40
42
44

32
36
40

Q26. What will be the output of the following pseudocode?

```
Integer p, q, r
Set p =0, q=7, r=5
if((p+q)>(r-p) OR (r<q)
  q=(9+2)^q
  r=q+r
End if
Print p+q+r
```

27

38

35

29

Q27. What will be the output of the following pseudocode?

```
Integer a,b,c
Set a=2,b=2,c=15
if (a<b && (c+a)>(a-c))
   c=c+a
End if
```

```
6 Print a+b+c
```

- 39
- 25
- 5
- 19

Q28. What will be the output of the following pseudocode?

```
1 Integer p, q, r
2 set p= 1, q= 1, r= 3
3 if (p && q>1 && r>5)
4    p = p + 1
5 end if
6 q = q + 1
7 printf p + q + r
```

- 12
- 6
- 17
- -10

Q29. What will be the output of the following pseudocode?

```
1 Integer a,b,c
2 Set a=1,b=4,c=8
3 if (b>c OR (a+b) < (b-a))
4    a=(7+10)+a
5    b=(8+5)+a
6 End if
7 Print a+b+c
```

- 8
- 17
- 26
- 13

Q30. What will be the output of the following pseudocode?

```
1 char str[20]
```

```
2  Integer s
3  Set str = " ABCDEFGH "
4  s = string_length(str)
5  str[5] = NULL
6  s = strlen(str)
7  print s
```

- 10
- 6
- 4
- 5

Answer Key & Solution

Section 1 - Aptitude

Q1

16

Solution

3664XX is divisible by 36 means it is divisible by both 4 and 9.

Last two digits should be divisible by 4.So it could be either 44 or 88

Sum of the digits should be divisible by 9 .

If X=4, (3+6+6+4+4+4= 27 divisible by 9)

If X=8, (3+6+6+4+8+8= 35 which is not divisible by 9)

X= 4. X^2 = 16

Q2

243

Solution

The base of all the numbers is same and only the powers are different.

The required H.C.F is the number with minimum power = 3^5 = 243

Q3

2166.67

Solution

Now, in this numerical, there is no common loss and gain %.

Hence, solve it making equations.

Let cost price of camel be x.

As cost of camel and carriage = Rs 5000

Cost of carriage = Rs. (5000 – x)

After selling camel he gains 20% and on carriage a loss of 10%. But on the whole he gains 3%.

Therefore,

20% of x – 10 % of (5000 – x) = 3 % of 5000

2x-5000+x=1500

3x=1500+5000

x=2166.67

The cost of camel = Rs. 2166.67

Q4

42.22%

Solution

Let CP be Rs. 3600 and MP be 100x

So, SP = 60x and Profit = Rs. 1200

So, SP = 3600 + 1200 = 60x

x = 80

So, discount = 40x = 40 * 80 = 3200

New discount = 3200 * 0.9 = Rs. 2880

So, New SP = 8000 - 2880 = Rs.5120

Hence, Profit % = (5120 - 3600)/3600 * 100% = 42.22%

Q5

900

Solution

Let 1% =1

Then, 10% = 10, 15% = 15, and 20% = 20

LCM of 10, 15, and 20 is 60.

The ratio of the incomes of Guru: Raj: Kamal = 6:4:3

Income of kamal = 3900/13 * 3 = 900

Q6

12000

Solution

Winning candidate got= 66% votes

Losing candidate got= 34% votes

Difference in votes= 66-34= 32% votes, which is equal to 3072 votes.

No of casted votes= 3072x100/32= 9600 votes

Since, 80% valid votes= 9600

total votes= 9600x100/80= 12000 votes.

Q7

Rs. 4000

Solution

Excess money obtained = 3 years @ 4% per annum

= 12% of Principal

So, according to the question, 12% = Rs. 480

So, 100% = Rs. 4000

Q8

4%

Solution

Let the principal be Rs x.

Then the SI =4/25x

Rate of interest = Time

r= (100*4/25x)/ x*r

r^2 =400/25

r=20/5=4%

Q9

9 hours

Solution

From given data,efficiency of men equal to women

No.of hours worked by men =90x8=720

M1*D1*H1=M2*D2*H2

30*90*8=24*100*y

y=9 hours

Q10

16 days

Solution

$3m + 4b = \left|\frac{2100}{7}\right| = 300$ per day ...(i)

$11m + 13b = \frac{8300}{8} = 1037.5$ per day ...(ii)

eq^n (i) × 11 — eq^n (ii) × 3

$11 \times 4b - 13 \times 3b = 11 \times 300 - 3 \times 1037.5$

$5b = 187.5$

$\therefore b = Rs.\ 37.5$

Again, $3m + 4 \times 37.5 = 300$

$3m = 300 - 150 = 150$

$\therefore m = Rs.\ 50$

Now, the required number of days

$= \frac{11000}{7 \times 50 + 9 \times 37.5} = \frac{11000}{687.5} = 16$

Q11

30km/hr

Solution

Total distance = 120 + 180 = 300 km.

Time taken for the first 120 km = 120/20 → 6 hrs.

Time taken for the next 180 km = 180/45 → 4 hrs.

Total time taken = 6 + 4 = 10 hrs.

Average Speed =Total Distance Travelled/Total Time Taken

= 300/10 = 30 km/hr.

Q12

1.25 km

Solution

By applying proportionality

Vinu Car

30m 48m

? 2000m

= 30/48 *(2000) = 1250m

Q13

9

Solution

Let the number of field workers be x

There are 16 documentation workers

Given total average = 45

Total age of all workers/ (16 + x) = 45

Total age of all the workers = 45(16 + x)

The average age of 16 documentation workers = 38

⇒ Total age of documentation workers = 38 × 16

⇒ Total age of field workers = x × 52

Total age of all the workers = total age of field workers + total age of documentation workers

⇒ 45(16 + x) = x × 52 + 38 × 16

⇒ 45 × 16 + 45x = 52x + 38 × 16

⇒ 720 − 608 = 7x

⇒ 7x = 112

⇒ x = 16∴ Unmarried field workers = (16 − 7) = 9

Q14

188

Solution

avg of 6 numbers = 78

therefore total sum of 6 nos = 6 * 78 = 468

avg of 1st five numbers = 56

therefore sum of first five numbers = 5 * 56 = 280

sixth no = 468 - 280 = 188

Q15

30000

Solution

Ratio of Capitals = X : 20000

3:2 = X:20000

X = 30000

Q16

16

Solution

Let the present age of kunal and Sagar be x and y years

Six years ago

$(x - 6)/(y - 6) = 6/5$

$5x - 6y + 6 = 0$ ----(1)

Four years hence

$(x + 4)/(y + 4) = 11/10$

$10x - 11y = 4$ ----(2)

From (1) and (2), we get x = 18 and y = 16

Sagar's present age = 16 years

Q17

450

Solution

Let,

e=No. of engineering students, m = no. of MBA students and c = no. of CA students

$4e + 3m + 5c = 3650$, therefore $e = (3650-3m-5c)/4$ --------(1)

$3c = 2m$, therefore $c = 2m/3$

$3e = 2c$ replacing e with (1) we get,

$3(3650- 3m - 5c)/4 = 2c$.....replacing c with 2m/3 and solving this equation we get m

= 450

So the number of MBA students is 450

Q18

3600

Solution

Total number of arrangements = (N-1)! ie 7!

If 2 brothers are together, then number of ways = 6! × 2!

Required number of ways Two brothers are not together = 7! – 6! 2! = 5040 – 1440 = 3600.

Q19 1/20

Solution

No. of tickets numbered such that it is divisible by 20 are 10000/20 = 500.

Hence required probability = 500/10000 = 1/20.

Q20 11/36

Solution

Probability of not getting 3 = $\frac{5}{6} \times \frac{5}{6} = \frac{25}{36}$

Probability of getting at least one 3 = $1 - \frac{25}{36} = \frac{11}{36}$

Q21 32589

Solution

Series Pattern Given Series

1 1

1 + 2 = 3 3

3 + 2 × 3 = 9 9

9 + 2 × 3 × 5 = 39 39

39 + 2 × 3 × 5 × 7 = 249 249

249 + 2 × 3 × 5 × 7 × 11 = 2559 2559

2559 + 2 × 3 × 5 × 7 × 11 × 13 = 32589 32589

Q22 222

Solution

Series Pattern	Given Series
$1^3 + 1 = 2$	2
$2^3 + 2 = 10$	10
$3^3 + 3 = 30$	30
$4^3 + 4 = 68$	68
$5^3 + 5 = 130$	130
$6^3 + 6 = 222$	**222**

Q23 288

Solution

Series Pattern Given Series

9 × 61 = 54 54

9 × 52 = 225 225

9 × 43 = 576 576

9 × 34 = 729 729

9 × 25 = 288 288

9 × 16 = 9 9

Q24 Sister

Solution

The only son of the woman's grandfather is the Woman's father. Man's brother's father is Man's father. So. the woman is the man's sister.

Q25 Sister-in-law

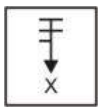
Solution

Martin’s daughter’s mother — Martin’s wife;

Martin’s wife’s father — Martin's father-in-law;

Father-in-law’s daughter — Martin’s sister-in-law or wife.

So, Hannah is Martin’s sister-in-law.

Q26 

Solution

The arrow rotates through 90° clockwise in such a way that one line segment from the arrow is removed and other sign rotates through 45°

Q27



Solution

The main figure rotate through 180° and the head of the arrow is reversed.

Q28

if only conclusion 2 is true

Solution

Combining the inequalities;

X≤Y<W =Z ≤U<S>T ≥V

No relationship can be established between Z and T

But we can clearly say that Z > X;

Hence II is true.

Q29

if only conclusion II is true

Solution

Combining the two equations, L ≥ M > N ≥ P M > P thus, I is false L ≥ M > N ≥ P L > N. Thus, II is true

Q30

if the question can be answered by using both the statements together, but cannot be answered by using either statement alone.

Solution

Statement (A) gives that A alone can produce 50 widgets in one day and 250 widgets in 5 days, It doesn't give any information about B.

So, statement (A) alone is not sufficient.

Similarly statement (B) alone is not sufficient.

Combining statement (A) and (B) we can find number of days required by machines A and B to produce 500 widgets.

Q31

If the data in both statements I and II together are necessary to answer the question

Solution

The common word between the question and I is they, which is nop in the code.

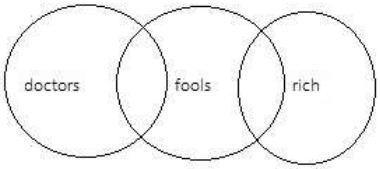
The common word between the question and II is like, which is all in the code.

So, ed will have to be flowers. Thus both the statements are needed to answer the question.

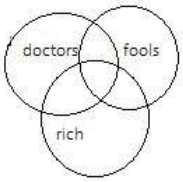
Q32

If neither I nor II follows.

Solution



OR

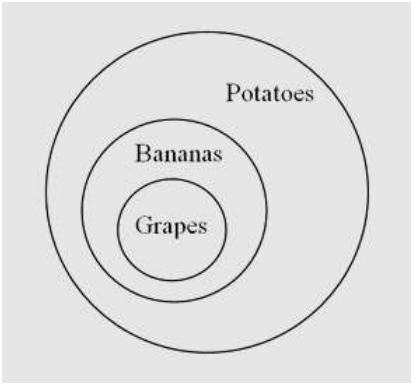


Neither conclusion follows

Q33

if both conclusions I & II follow.

Solution



Both I & II follow. So option 3

Q34

a

Solution

Here, for the given blank 'a' suits the best.

Since, we talk about a brief period of time (really good time), we select 'a' as the answer.

Q35

at

Solution

"at" is used to identify the object of a sentence.

Q36

In , for

Solution

In general, "in" is used to indicate location or position within or inside something

"For" is used to introduce an object

Q37

A

Solution

'The creature on Mars, if any,' in the sentence should be replaced by 'the creatures on Mars, if any'.

The plural of 'creatures is used to agree with the verb 'are' and to correspond with 'if any'. A verb must agree with its subject in number and person. The plural 'parts' fits in with the verb 'are'.

Q38

3

Solution

Ans (3). Error is in part 3. 'Anybody' is an indefinite pronoun and conveys singular meaning contextually. So the antecedent should also be a singular pronoun. Hence replace 'they' using 'he/she' to make it as a grammatically correct sentence.

Q39

ignored

Solution

Ans. (1). In the context of Nirmal Verma's stories being 'read too narrowly' and '...Premchand's social 'reformer role ...' the .word 'ignored' (1) is the right word. Hence, option (1) is the right option.

Q40

Combination, dent

Solution

The first blank should be "combination", since the first part of the sentence talks about two different things.

Also, the second blank should take "dent" which means to slow down, or resist

Q41

BDAC

Solution

Ans. (3). Statement B initiates the paragraph. D followed by A furthers the argument, by telling us that even though the projections show a near- doubling in the next two years, hardly any benefit of the offshore drilling activity in the state has been received by any of the residents. A researcher comments that the oil revenues have not been directed towards progress of the country in the concluding statement C. Option (1): C and A though giving the same indication are separated by "Yet" which is contradictory. Option (2): D, which is a specific statement, precedes B, which is a general statement on the situation and should pave the way for the specifics, Option (4): Again, "Yet" of statement A contradicts the similar tone of C and A. Hence, the correct order is BDAC and the answer is (3).

Q42

(c) BDACE

Solution

From the options it is clear that Statement D and Statement C cannot be a starting statement because statement D is like sentence continuation and C starts with 'the' which is used to re-refer the object.

So, We can eliminate option (a) and option (d).

Now, We can go with option (b) and option (c).

From this we can confirm that B is the starting statement. Followed by Statement D as it is the perfect continuation.

With all those information we can include option (c) is the answer.

Q43

did you take

Solution

Here yesterday indicates the statement is in Past tense

Hence, the only past tense option here is DID YOU TAKE

Q44

forget

Solution

The verb "forget" is used here to indicate the action of failing to remember to keep the keys in the correct place.

Q45

This game will never be played by me.

Solution

Solution:The given sentence is in simple future tense and is in active voice.

When the voice is changed, will becomes will be

The right answer is "This game will never be played by me."

Q46

He told me that he had been working for the same company since 1960.

Solution

Solution:

Said is change to told. have been becomes had been. The right answer is 'He told me that he had been working for the same company since 1960.'

Q47 rustic

Solution

The word yokel means bumpkin or clown. Ignoramus refers to known nothing, dandy refers to fine and recluse refers to outsider. So the best answer is rustic which means rural or pastoral.

Q48 prudent

Solution

The word judicious means careful or cautious.

Fair refers to reasonable, honest refers to truthful and loyal refers to faithful.

So the best answer is prudent which means careful or cautious.

Q49. Grandson

Solution

G and D are the children of B while F is the father of D. Hence, F must be the husband of B.

Now as one of G and D is male and the other is female and since H is the husband of G, G must be a female and D is one of her two brothers.

Further, it's given that C is the daughter-in-law of G. Hence, she has to be married to the son of G.

Since B has only one grandchild i.e. E, E has to be the son of G and H.

Thus, F-B, H-G and E-C form three couples.

Now, since D is one brother of G, the only person remaining i.e. A, therefore, has to be the other brother.

Q50. C is the granddaughter in law of F.

lution

and D are the children of B while F is the father of D. Hence, F must be the husband of

.

ow as one of G and D is male and the other is female and since H is the husband of G,

must be a female and D is one of her two brothers.

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hus, F-B, H-G and E-C form three couples.

ow, since D is one brother of G, the only person remaining i.e. A, therefore, has to be the

ther brother.

Q51. T

Solution

School	Day	Student
St.Joseph	Monday	B
Kendriya Vidyalaya	Tuesday	T
Sri Amrita	Wednesday	J
Cauvery Global	Thursday	S
Nava Bharath	Friday	P
St.Anne's	Saturday	D
Bala Vidya Mandhir	Sunday	H

Q52. Nava Bharat – Friday

Solution

No Solution

Q53. St.Joseph's- B

Solution

No Solution

Q54. Kendriya Vidyalaya

Solution

School	Day	Student
St.Joseph	Monday	B
Kendriya Vidyalaya	Tuesday	T
Sri Amrita	Wednesday	J
Cauvery Global	Thursday	S
Nava Bharath	Friday	P
St.Anne's	Saturday	D
Bala Vidya Mandhir	Sunday	H

Q55. J

Solution

School	Day	Student
St.Joseph	Monday	B
Kendriya Vidyalaya	Tuesday	T
Sri Amrita	Wednesday	J
Cauvery Global	Thursday	S
Nava Bharath	Friday	P
St.Anne's	Saturday	D
Bala Vidya Mandhir	Sunday	H

Q56. the principle that people's happiness is the Kings's happiness.

Solution

The tenth paragraph, in the opening line, states and upholds option (the principle that people's happiness is the Kings's happiness.) as the correct one. Other options are not correct in the given context;

Hence, option (3the principle that people's happiness is the Kings's happiness.) is the correct option.

Q57. must be free from anger and hatred.

Solution

The third paragraph mentions the option "must be free from anger and hatred".

The passage also says 'For punishment, when awarded with due consideration" which means there should be a valid reason which is not influenced by the self motto. The other options are mentioned in the passage, but are not the correct answers in the given context.

Q58. the sanctity of life.

lution

he first paragraph states and upholds that the principles of good governance should be stablished in the sacredness of life. This is reflected in option (2).

ptions (1), (3) and (4) are incorrect. Hence option (2) is the correct option.

Q59. the foundations of all relationships are in knowing the sanctity of life.

Solution

The third paragraph states and upholds option (the foundations of all relationships are in knowing the sanctity of life.), which is the correct one. Other options are incomplete and are not as strong as option (the foundations of all relationships are in knowing the sanctity of life.).

Hence option (the foundations of all relationships are in knowing the sanctity of life.) is the correct option.

Q60. wrath

Solution

The sixth paragraph towards the end endorses the option "wrath".

It is the option sought. Wrath means extreme anger.

Section 2 - Pseudocode and Computer Fundamentals

Q1 RISC

Solution

Explanation: Hence the RISC architecture is followed in the design of mobile devices.

Q2 Computer Speed Addition

Solution

Explanation: The CSA is used to speed up the addition of multiplicands.

Q3 Memory data register

Solution

No Solution

Q4 Source Program

Solution

Explanation: The input we give in a high-level language is also known as the source language.

Q5 byte

Solution

No Solution

Q6 Half-duplex mode

Solution

No Solution

Q7 Software available on hardware

Solution

No Solution

Q8 Both Magnetic tapes, disks and Magnetic bubbles

Solution

No Solution

Q9 Modifying or adding records

Solution

No Solution

Q10 Electrically Erasable Programmable Read-Only Memory

Solution

No Solution

Q11 Register

Solution

No Solution

Q12 thick and thin lines

Solution

No Solution

Q13 Spreadsheet program

Solution

No Solution

Q14 Incorrect data entry.

Solution

No Solution

Q15 SNARC

Solution

No Solution

Q16 33

Solution

No Solution

Q17 67

Solution

a=4,b=12,c=41, d=0

Inside loop:

d=0+4+12+41 = 57

e=57/3 = 19

a=4+5 = 9

b= 12+5 = 17

Same process repeated until condition gets failed. Updated value is a=19,b=27,d=201,e=67. Finally 67 gets printed.

Q18 54

Solution

Inside for loop, initially j=0,array1 [0]%3 =0 (True) so it executes if block and update q and p value. The same process is repeated until j value reaches 5.

While j=0 , p= -6, q= -6

While j=1 , p= 24, q= 27

While j=2 , p= 24, q= 31

While j=3 , p= 24, q= 33

While j=4 , p= 24, q= 44

While j=5 , p= 24, q= 54 (then loop terminates and print q value as 54)

Q19

1

3

Solution

In fun(1): print 1, then 1< 3 (true), c=1+2=3, fun(fun(3)) (nested recursion) .

In fun(3): print 3, then 3<3 (false), return 3 to fun(1) and again nested recursion calls and print 3. So 1 3 gets printed.

Q20

2

Solution

Initially a[4] = { 1, 1, 1, 2}

a[2]= 1 - 1 + 2 => a[2]=2

a[3] = 2

m = 2

Q21

44

Solution

No Solution

Q22

26

Solution

Initially a=5,b=7,c=7. When c=3,if block gets execute and update a=6,b=4 and outside if block again 'a' updated to 10.

When c=4,if block gets execute and update a=8,b=4 and outside if block again 'a' updated to 4.

When c=5,if block gets execute and update a=10,b=9 and outside if block again 'a' updated to 17.Then loop terminates and add 17+9=>26 gets printed.

Q23

104

Solution

Initially p=7, q=8, r=7,then by applying bitwise operator values are updated to r=12 ,p=32, q=0. If block condition gets true so again p value update to p=68 inside if block. Finally outside if block also p value updated to 92. So 92+12+0 => 104 gets printed.

Q24

15

Solution

In funn(4 2) => return 1+funn(14,14)

In funn(14,14) => return 14 to funn(4 2)

In funn(4 2) => return 1+14 => 15 gets printed.

Q25

42

46

48

Solution

When i=1, updated value of a and b is 42,9.

When i=2, updated value of a and b is 46,4.

When i=3, updated value of a and b is 48,3.

So 42 46 48 gets printed.

Q26

29

Solution

If block gets executed and updated q and r value as 12 and 17. So 0 +12+17=29.

Q27

19

Solution

If condition gets failed so 2+2+15 = 19

Q28

6

Solution

If the condition gets failed so only the q value will be updated.

So, p=1, q=2, and r=3.

As a result, 3 will be printed.

Q29

13

Solution

If block condition gets failed so 1+4+8=13.

Q30

5

Solution

Null is considered as the end of the string.